Bài tập tính toán

Thứ ngày . .

Bài 1

$$q = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$S_1 = q^T k_1 = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

$$S_2 = q^T k_2 = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2$$

$$S_3 = q^T k_3 = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3$$

Bài 2 - Softmax

$$S = [1, 2, 3]$$

$$\alpha_1 = \frac{e^1}{e^1 + e^2 + e^3} = 0,09$$

$$\alpha_2 = \frac{e^2}{e^1 + e^2 + e^3} = 0,2447$$

$$\alpha_3 = \frac{e^3}{e^1 + e^2 + e^3} = 0,6652$$

$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = 1$

Vì e^x là hàm đồng biến $\Rightarrow$ Score càng lớn thì e^x càng lớ

Bài 3

$$\alpha = [0,1 ; 0,2 ; 0,7]$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

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$C = 0,1 v_1 + 0,2 v_2 + 0,7 v_3$

$= 0,1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0,7 \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$= \begin{bmatrix} 1,5 \\ 1,6 \end{bmatrix}$

v_3 đóng góp nhiều nhất vì có attention weight lớn nhất $\alpha_3 = 0,7$

Bài 4

$s_1 = q^T k_1 = [1\ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$

$s_2 = q^T k_2 = [1\ 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$ $\Rightarrow s = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$s_3 = q^T k_3 = (1\ 0) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1$

$\alpha_1 = \frac{e^1}{e^1 + e^0 + e^1} = 0,4223$

$\alpha_2 = \frac{e^0}{e^1 + e^0 + e^1} = 0,1554$

$\alpha_3 = \frac{e^1}{e^1 + e^0 + e^1} = 0,4223$

$C = \alpha_1 v_1 + \alpha_1 v_2 + \alpha_3 v_3$

$= 0,423 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0,1559 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 0,4223 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1,2869 \\ 0,7331 \end{bmatrix}$

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Thứ ngày . .

1 Scaling

$$\frac{QK^T}{\sqrt{4}} = [2, 1, 0]$$

$$\alpha_1 = \frac{e^2}{e^2+e^1+e^0} = 0,6652$$

$$\alpha_2 = \frac{e^1}{e^2+e^1+e^0} = 0,2447$$

$$\alpha = \frac{e^0}{e^2+e^1+e^0} = 0,09$$

TH phóng scaling

$$\alpha_1 = \frac{e^4}{e^4+e^2+e^0} = 0,866$$

$$\alpha_2 = \frac{e^2}{e^4+e^2+e^0} = 0,1172$$

$$\alpha_3 = \frac{e^0}{e^4+e^2+e^0} = 0,0159$$

2 1 Self attention

["I", "like", "cats"]

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}; \quad W_V = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$Q = XW_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$K = XW_K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}$$

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$$V = XW_V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{matrix} \text{Thứ} \\ \text{ngày} \end{matrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$S = \frac{QK^T}{\sqrt{d_k}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 0,7071 & 0 & 0,7071 \\ 0,7071 & 0,7071 & 1,4142 \\ 1,4142 & 0,7071 & 2,1213 \end{bmatrix}$$

Ta có $a_{ij} = \frac{e^{S_{ij}}}{\sum_k e^{S_{ik}}}$

$$\Rightarrow A = \begin{bmatrix} 0,4011 & 0,1978 & 0,4011 \\ 0,2483 & 0,2483 & 0,5035 \\ 0,2840 & 0,14 & 0,576 \end{bmatrix}$$

$$C = A \cdot V$$

$$= \begin{bmatrix} 0,8022 & 1,1978 \\ 0,7517 & 1,5035 \\ 0,86 & 1,432 \end{bmatrix}$$

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